APEXPLANET DATA ANALYTICS INTERNSHIP

Data Storytelling & Statistical Validation

Task 4 | E-commerce Sales Analysis

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A business-focused summary of sales analysis and a statistical test comparing average sales by gender.

1. Business Objective & Research Question

Business objective

Investigate whether average sales differ significantly between male and female customers in the available e-commerce dataset.

Research question

Is there a statistically significant difference in average sales between male and female customers?

NULL HYPOTHESIS (H■)	ALTERNATIVE HYPOTHESIS (H■)
There is no statistically significant difference in average sales between male and female customers.	There is a statistically significant difference in average sales between male and female customers.

Significance level: $\alpha = 0.05$ (5%)

2. Dataset Overview

The analysis uses the cleaned dataset prepared during Task 1.

1,000	13	489	511
Rows / records	Columns / fields	Female observations	Male observations

Key fields

Gender, Total_Sales, Order_ID, Order_Date, Customer_ID, City, Product, Category, Quantity, Unit_Price and Sales_Outlier.

Preparation

The cleaned CSV was loaded into Python with pandas. Sales were grouped by Gender and summarized using count, mean, median and sum.

3. Descriptive Analysis: Sales by Gender

Male customers have a higher observed mean and total sales in this dataset. Descriptive differences alone do not prove a statistically significant effect.

Metric	Female	Male
Observations	489	511
Mean sales	136,883.21	141,807.34
Median sales	108,470.60	109,237.86
Total sales	66,935,889.02	72,463,550.63

Observed mean difference (Male – Female): 4,924.13

Currency denomination was not specified in the dataset, so values are shown without a currency symbol.

4. Statistical Method

Test used: Independent two-sample Welch’s t-test.

This test compares the means of two independent groups and does not require equal population variances.

1	**Define groups** Female sales and male sales.
2	**Set decision threshold** Use $\alpha = 0.05$.
3	**Run the test** Calculate the t-statistic and p-value.
4	**Interpret** Compare p-value with α and examine the 95% confidence interval.

The test evaluates evidence for a difference in population means; it does not establish causation.

5. Hypothesis Test Results

T-statistic	P-value	95% confidence interval for mean difference
0.6826	**0.4950**	–9,231.58 to 19,079.85

Decision: FAIL TO REJECT H■

The p-value (0.4950) is greater than 0.05, and the 95% confidence interval includes zero. The analysis does not provide sufficient evidence of a statistically significant difference in average sales between male and female customers.

This does not prove the group means are exactly equal; it means a difference was not detected at the chosen significance level.

6. Business Recommendations

Based on this test, gender alone should not be used as the basis for a sales strategy.

Use multiple signals	Evaluate product category, city, order value, purchase frequency and customer recency alongside gender.
Test campaigns	Run controlled marketing experiments and compare conversion, average order value and revenue.
Segment customers	Create segments based on purchase behavior and engagement rather than assuming gender explains spending.
Track outcomes	Monitor campaign performance over time and validate decisions with data.

These are suggested actions based on the statistical finding and should be validated against business goals and further analysis.

7. Conclusion & Call to Action

KEY FINDING

Observed male mean sales were higher, but the difference was not statistically significant at $\alpha = 0.05$.

Call to action

Continue evaluating customer behavior, test marketing ideas with controlled experiments, and make decisions using statistically validated evidence.

Thank you

Task 4 — Data Storytelling & Statistical Validation

Before submitting, add relevant charts and findings from Tasks 1–3 if required by the internship brief. Verify the organization name and dataset units.